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High Tech
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DEPARTMENT OF COMPUTER SCIENCE ENGINEERING ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

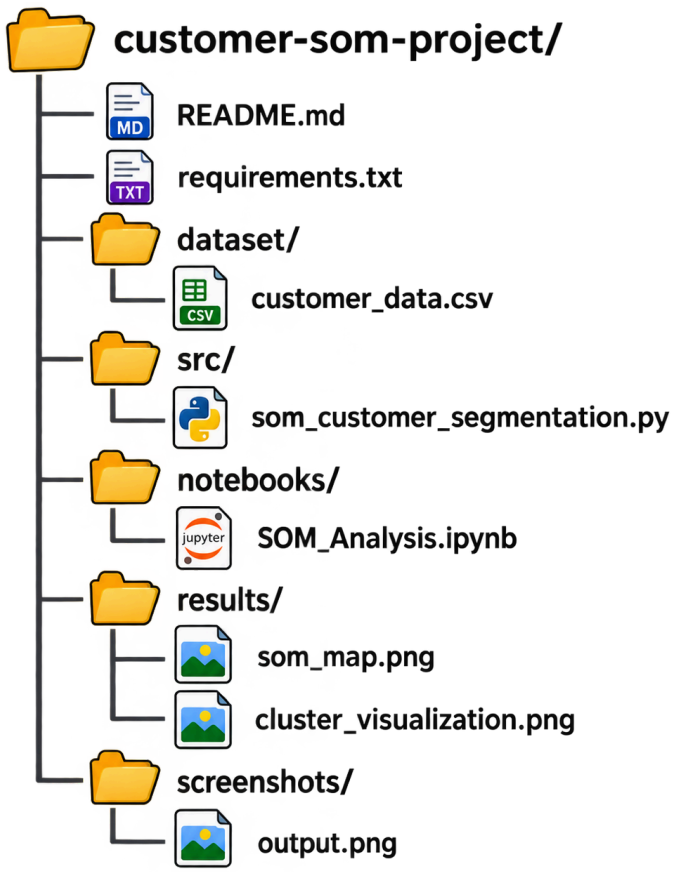
INNOVATIVE ASSIGNMENT – II

FACULTY NAME:	**N SHARMILA BANU**	**FACULTY ID:**	**HTS 1787**
COURSE CODE:	**21ML53T-26**	**COURSE NAME:**	**DEEP LEARNING ESSENTIALS**
YEAR/SEM	**III/V**	**STUDENTS STRENGTH / BATCHES /SEC**	**62 / 20 (B Sec)**

Rules & Authenticity Verification

1. Individual work only.
2. Create and submit your own GitHub repository.
3. Maintain meaningful Git commits during development.
4. Code must be understood and explainable by the student.
5. Include README, source code, and output/results.
6. External resources/code must be properly acknowledged.
7. The faculty will verify Commit History and Git Blame.
8. Students must demonstrate the project and answer viva questions.
9. Faculty may ask for a live modification to verify authenticity.
10. Copied/unexplained projects will not be considered authentic.

GitHub Repository Structure



REPORT STRUCTURE

1. Title Page
2. Problem Statement
3. Objective
4. Concept Used
5. Methodology / Working Steps
6. Implementation
    - Tools & Libraries
    - Source Code / GitHub Repository
7. Results & Output
8. Analysis
9. Conclusion
10. References

INNOVATIVE ASSIGNMENT-II PROBLEM STATEMENTS

BATCH NO	STUDENT NAME	PROBLEM STATEMENT	CO	Level
1	MOHAN KUMAR M	**Topic:Edge Detection Using Basic Convolution** Develop a small image-processing application that applies horizontal and vertical convolution filters to a grayscale image to identify and highlight edges.	CO3	K3
	SESSHANTH L S	**Topic: Transfer Learning for Flower Image Classification** Use a pre-trained CNN model and fine-tune its final layers to classify different types of flowers from a small custom dataset.	CO4	K4
	MONICA A	**Topic: Image Reconstruction Using Autoencoders** Develop a basic autoencoder that compresses input images into a latent representation and reconstructs the original images from the compressed information.	CO5	K4

BATCH NO	STUDENT NAME	PROBLEM STATEMENT	CO	Level
2	MONIKA D	**Topic: Transfer Learning for Animal Recognition** Develop an image classification system using a pre-trained CNN model to recognize a limited set of animal categories with a small training dataset.	CO4	K3
	MONISHA S	**Topic: Image Sharpening Using Convolution Kernels** Design an application that enhances blurred images using different sharpening convolution kernels and compares the resulting image quality.	CO3	K4
	NARENDHIRAN R	**Topic: Latent Space Visualization Using Autoencoders** Build an autoencoder and visualize its latent-space representations to understand how similar images are grouped in the compressed feature space	CO5	K4

BATCH NO	STUDENT NAME	PROBLEM STATEMENT	CO	Level
3	NISHANT S	**Topic:Image Noise Reduction Using Convolution** Develop a program that reduces noise from digital images using average and Gaussian-style convolution filters and evaluates the output.	CO3	K3
	RAVINDRA S PARINITHA MUGTHA KAVYA	**Topic:Transfer Learning with Feature Extraction** Use a pre-trained CNN as a feature extractor and train a simple classifier on the extracted features for a new image classification task.	CO4	K4

	R	**Topic: Image Denoising Using Autoencoders** Develop a denoising autoencoder that takes noisy images as input and reconstructs cleaner versions of the original images**.**	CO5	K4

BATCH NO	STUDENT NAME	PROBLEM STATEMENT	CO	Level
4	PARTHASARATHI M	**Topic: RGB Multi-Channel Convolution** Implement convolution separately on the Red, Green, and Blue channels of an image and combine the resulting feature maps.	CO3	K3
	PAVITHRAN M	**Topic: Transfer Learning with Layer Freezing** Compare a transfer-learning model with frozen convolutional layers against a model where selected layers are fine-tuned for a custom image dataset.	CO4	K4
	YOKESHWARAN K	**Topic: Image Compression Using Autoencoders** Create an autoencoder-based image compression system that reduces image dimensions through the encoder and reconstructs the image using the decoder.	CO5	K4

BATCH NO	STUDENT NAME	PROBLEM STATEMENT	CO	Level
5	PRAVEENGANDHI V	**Topic: DenseNet-Based Image Classification** Implement a lightweight DenseNet model to classify images into multiple categories and analyze the classification performance.	CO4	K3
	PRIYADHARSHINI G S	**Topic: Stride and Padding Visualization** Create a visualization tool that demonstrates how different stride and padding values change the size and structure of convolution outputs.	CO3	K4
	PULIVARTHI AKSHAYA CHOWDARY	**Topic: Anomaly Detection Using Autoencoders** Train an autoencoder using normal data and identify unusual samples based on their reconstruction error.	CO5	K4

BATCH NO	STUDENT NAME	PROBLEM STATEMENT	CO	Level
6	RAGHA PRIYAN R	**Topic: Convolution Output Shape Calculator** Develop a utility that calculates the output dimensions of a convolution layer based on input size, kernel size, stride, and padding.	CO3	K4

	RAGULKANTH K	**Topic: Dense Connectivity Visualization** Develop a visualization application that demonstrates how feature maps are passed between densely connected layers in a DenseNet architecture.	CO4	K4
	PANDARAM DILLIBABU	**Topic: Simple Deep Belief Network for Pattern Recognition** Implement a small Deep Belief Network to learn hierarchical representations from a simple image or pattern dataset and perform classification.	CO5	K3

BATCH NO	STUDENT NAME	PROBLEM STATEMENT	CO	Level
7	RITHIKAA T	**Topic: DenseNet-Based Medical Image Categorization** Apply a pre-trained DenseNet model to categorize a small collection of medical images into predefined classes.	CO4	K4
	ROGEETH E	**Topic: CNN Feature Map Generator** Build a simple CNN module that applies multiple filters to an input image and displays the feature maps generated by each filter.	CO3	K3
	ROSHAN ANTONY C	**Topic: Feature Learning Using Deep Belief Networks** Develop a Deep Belief Network that learns useful features from unlabeled input data and visualize the learned representations.	CO5	K4

BATCH NO	STUDENT NAME	PROBLEM STATEMENT	CO	Level
8	RUBESH GAALI RAVI	**Topic: Pixel-Level Image Classification Using PixelNet** Develop a PixelNet-based application that assigns class labels to individual pixels in an image for a simple scene or object segmentation task.	CO4	K4
	RUKESH S G	**Topic: Binary Pattern Learning Using Boltzmann Machines** Implement a small Boltzmann Machine to learn and reconstruct simple binary patterns such as digits, symbols, or geometric shapes.	CO5	K3
	SANDHIYA S	**Topic: Convolution with Different Data Types** Implement convolution using integer and floating-point data types and compare their computational behavior and output values.	CO3	K4

BATCH NO	STUDENT NAME	PROBLEM STATEMENT	CO	Level
9	SANDHYA S	**Topic: Low-Precision Convolution Analysis** Study the effect of reduced numerical precision on convolution results by comparing different precision levels in terms of accuracy and computation	CO3	K4
	SANJAY ANAND S	**Topic: PixelNet Feature Representation Analysis** Implement a simplified PixelNet architecture and visualize pixel-level feature representations generated from an input image.	CO4	K4
	SANJAY KRISHNA K	**Topic: Recommendation Pattern Learning Using Boltzmann Machines** Develop a simple Boltzmann Machine that learns user-item interaction patterns and predicts missing preferences from a small dataset.	CO5	K3

BATCH NO	STUDENT NAME	PROBLEM STATEMENT	CO	Level
10	SANJAY N	**Topic: Deep Feature Learning Using Deep Boltzmann Machines** Build a simplified Deep Boltzmann Machine with multiple hidden layers to learn hierarchical representations from a small dataset.	CO5	K4
	MOHANRAJ S	**Topic: Memory-Efficient Feature Extraction** Develop a convolution-based feature extraction system that reduces unnecessary intermediate feature-map storage and analyzes memory usage.	CO3	K4
	SANTHOSH M(27.12.2006)	**Topic: RNN-Based Sequence Prediction** Develop an RNN model that learns a short numerical sequence and predicts its next value based on previously observed values.	CO4	K3

BATCH NO	STUDENT NAME	PROBLEM STATEMENT	CO	Level
11	SANTHOSH M(30.12.2006)	**Topic: Fast Convolution Using Matrix Multiplication** Implement convolution by transforming the input into a matrix representation and performing matrix multiplication. Compare its execution time with direct convolution.	CO3	K4
	SASIKALA G	**Topic: RNN-Based Text Character Prediction** Create an RNN application that predicts the next character in a word or short text sequence based on previously entered characters.	CO4	K4

	SATHANA J R	**Topic: Image Generation Using a Basic GAN** Develop a basic Generative Adversarial Network consisting of a generator and discriminator to generate synthetic images resembling a selected training dataset.	CO5	K3

BATCH NO	STUDENT NAME	PROBLEM STATEMENT	CO	Level
12	SANJEEV S	**Topic: Mini ResNet Image Classifier** Develop a simplified ResNet model using residual connections to classify images into a small number of predefined categories.	CO3	K4
	SHALINI F	**Topic: RNN-Based Sentiment Sequence Classification** Build an RNN model that classifies short text reviews into predefined sentiment categories using sequential word information.	CO4	K4
	SHARUKA K	**Topic: GAN-Based Handwritten Digit Generation** Train a GAN to generate new handwritten digit images that resemble samples from a standard handwritten-digit dataset.	CO5	K4
	SELVARAJ	**Topic:Synthetic Image Dataset Generator** Create a GAN application that generates additional synthetic images to expand a small image dataset for machine-learning experiments.	CO5	K4

BATCH NO	STUDENT NAME	PROBLEM STATEMENT	CO	Level
13	SHRI CHARAN L	**Topic: GAN Training Progress Visualizer** Create an application that records and displays generated images at different stages of GAN training to observe how the generator improves its outputs.	CO5	K4
	SHYAM S	**Topic:Separable Convolution Implementation** Implement spatially separable convolution and compare its computational requirements with conventional two-dimensional convolution.	CO3	K4
	SIDDHARTH J	**Topic: Bidirectional RNN for Text Classification** Develop a Bidirectional RNN that processes text sequences in both forward and backward directions to classify short text samples.	CO4	K3

BATCH NO	STUDENT NAME	PROBLEM STATEMENT	CO	Level
14	SIRANJEEVI K	**Topic: Residual Connection Performance Study** Create two small CNN models, one with residual connections and one without, and compare their training performance and classification results.	CO3	K3
	SIVANESAN K	**Topic: Bidirectional RNN for Named Entity Identification** Implement a Bidirectional RNN to identify simple entities such as names, locations, or organizations in short sentences.	CO4	K4
	YUVARAJ T J	**Topic: Synthetic Tabular Data Generation Using GANs** Develop a GAN-based system that generates synthetic tabular records resembling a small real-world dataset while preserving the general statistical patterns.	CO5	K4

BATCH NO	STUDENT NAME	PROBLEM STATEMENT	CO	Level
15	NAKKA SRI VARSHITHA	**Topic: ResNet-Based Object Recognition** Develop an object recognition application using a lightweight ResNet model to classify objects belonging to selected categories.	CO3	K4
	SUJA SRI V	**Topic: LSTM-Based Time-Series Prediction** Develop an LSTM model that predicts future values from a small time-series dataset such as temperature, traffic, or energy consumption.	CO4	K3
	SUJEETH S	**Topic:Synthetic Image Dataset Generator** Create a GAN application that generates additional synthetic images to expand a small image dataset for machine-learning experiments.	CO5	K4

BATCH NO	STUDENT NAME	PROBLEM STATEMENT	CO	Level
16	SURYA E	**Topic: Mini AlexNet Image Classifier** Analyze the effect of different convolutional layers and hyperparameters on the classification performance and interpret the results using suitable evaluation metrics.	CO3	K4
	TAMIZH SELVAN R THARUN KUMAR R	**Topic: LSTM-Based Next Word Prediction** Analyze how different sequence lengths, LSTM configurations, and training parameters affect the accuracy of next-word prediction.	CO4	K4

		Topic: Conditional GAN for Digit Generation Analyze how different class labels, generator/discriminator configurations, and training parameters affect the quality and correctness of the generated digits.	CO5	K4

BATCH NO	STUDENT NAME	PROBLEM STATEMENT	CO	Level
17	THARUN Y	**Topic: LSTM-Based Activity Sequence Recognition** Develop an LSTM model that recognizes simple human activities from sequential sensor readings such as walking, standing, or sitting.	CO4	K3
	TIRUMURU MOURYA VARDHAN	**Topic: AlexNet Feature Map Visualization** Visualize the feature maps generated at different layers of an AlexNet-based CNN to understand how image features are extracted**.**	CO3	K4
	UDHAYAKUMAR G	**3.Topic: Conditional GAN for Object Image Generation** Develop a cGAN that generates images according to predefined object categories provided as conditional inputs.	CO5	K3
	SRINIVASAN	**Topic: PixelNet Feature Representation Analysis** Implement a simplified PixelNet architecture and visualize pixel-level feature representations generated from an input image.	C04	K4

BATCH NO	STUDENT NAME	PROBLEM STATEMENT	CO	Level
18	VAISHNAVI J	**Topic: AlexNet-Based Image Classification** Develop an image classification application using a pretrained or simplified AlexNet model for a selected set of image categories.	CO3	K3
	VEDHIKA M	**Topic:Vision Transformer for Image Classification** Implement a small Vision Transformer model to classify images into a limited number of categories and observe its classification performance.	CO4	K3
	VIJAY ANAND K	**Topic:Sketch-to-Image Translation Using GAN** Develop a GAN-based image translation system that converts simple sketches or edge maps into corresponding realistic-style images.	CO5	K3

BATCH NO	STUDENT NAME	PROBLEM STATEMENT	CO	Level
19	VIJAY BALAJI M	**Topic: Vision Transformer Patch Visualization** Analyze how different patch sizes affect the number of tokens, spatial information, and feature representation, and interpret the resulting visualizations.	CO4	K4
	VISHNU J	**Topic: CNN-Based Handwritten Digit Recognition** Develop a CNN application that recognizes handwritten digits from input images and displays the predicted numerical value.	CO3	K3
	PRADEEP RAJ J	**Topic: Day-to-Night Image Translation** Analyze how different model configurations and training parameters affect the visual quality and realism of the generated night-time images.	CO5	K4

BATCH NO	STUDENT NAME	PROBLEM STATEMENT	CO	Level
20	SORNA JEBA MARY J	**Topic: CNN–LSTM Hybrid for Video Activity Recognition** Analyze the contribution of spatial and temporal features to activity-recognition performance and interpret the model results	CO4	K4
	YUVASRI G	**Topic: CNN-Based Defect Detection** Analyze the effect of different CNN architectures, feature-extraction layers, or hyperparameters on defect-detection performance using suitable evaluation metrics.	CO3	K4
	HANNA FLORENCE E	**Topic: Colorization of Grayscale Images Using GANs** Develop a GAN-based system that converts grayscale images into color images by learning color patterns from a suitable training dataset.	CO5	K3

COURSE FACULTY **HOD** **DEAN ACADEMICS**